

- 7 (a) The function of the enzyme iduronate 2-sulfatase is to break down certain complex molecules.

Hunter syndrome is a rare, inherited condition in humans, caused by iduronate 2-sulfatase not functioning properly or by iduronate 2-sulfatase not being produced. This will cause the complex molecules to build up, leading to progressive damage to many organs.

Explain how a gene mutation can lead to iduronate 2-sulfatase not functioning properly or not being produced.

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- (b) Hunter syndrome is caused by a recessive allele located on the **X** chromosome.

Using appropriate symbols, draw a genetic diagram to show the probability of a normal male and a heterozygous normal female having a child with Hunter syndrome.

key

parental
phenotype

normal male

×

normal female

parental
genotype

gametes

offspring
genotypes

offspring
phenotypes

probability

[5]

[Total: 9]